

USER MANUAL

GRAVITY VERTICAL SYSTEM



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1. CERTIFICATION

The GVS system provides fall protection for persons climbing fixed ladders or similar structures and conforms to the EN353-1 standard: 2017.

The GVS system is designed to eliminate or substantially reduce a fall risk that would otherwise lead to serious injury or death of people working on elevated structures

2. PRELIMINARY CHECKS

It is mandatory to check the following before using the system:

2.1. User:

- The user of the lifeline must be in **good physical condition, and** not prone to vertigo or dizziness.
- User must have received adequate training including **(US22298) or (US2229998 & US2229995).**
- The total mass of each user, including **tools and equipment, must not exceed the** one indicated in the Fall Arrest System.
- A minimum of **2 persons** are required to work nearby, so that each can assist the other in the event of an incident.

2.2. Personal Protective Equipment (PPE):

- A suitable harness with two anchoring points or D rings, one at the back and one at the chest, in conformity with the EN 361 or ANSI / ASSEZ359.1
- The sternal anchorage is used to connect the glider.
- The fall arrest harness must be well positioned and tight to avoid hazardous **■** situations while the user is climbing up.

2.3. System:

- Check if the annual inspection was carried out within less than 1 year.
- There is no obstacle under the system that could reduce the fall clearance
- There are no deformed components
- **Rescue plan** must be in place to evacuate the person who fell within 15 minutes.
The rescue material must be easily accessible and be in the vicinity from the lifeline.

- The system must be immediately put out of service if the user notices one or more anomalies and the failing components must be replaced by a certified installer

3. USE OF THE SYSTEM

3.1. Before Climbing

- Inspect your full-body harness.
- Inspect the vertical lifeline system, including the cable, brackets, anchors, and glider.
- Ensure the system inspection is current and there is no visible damage.
- Verify the glider **is correctly orientated on the cable according** to the manufacturer's instructions.
- Connect the glider to the front fall-arrest attachment point on your harness using the approved connector.

3.2. Ascending the Vertical Lifeline

- Maintain three points of contact with the ladder where possible.
- Climb normally without pulling or forcing the glider.
- Allow the glider to follow your movement automatically.
- Pass intermediate brackets or anchors without disconnecting from the system (on systems designed for this purpose).
- Keep your body centred on the ladder and avoid excessive leaning.

3.3. Descending the Vertical Lifeline

- Face the ladder and descend in a controlled manner.
- Allow the glider to travel down the cable naturally as you move.
- Do not hold the glider open, override the locking mechanism, or pull it downward.
- Continue maintaining secure footing and three points of contact where possible.
- Once safely at ground level or the landing platform, disconnect only when you have reached a safe area and according to site procedures.

3.4. Common Mistakes to Avoid

- Climbing while disconnected from the glider.

- **Installing the glider upside down.**
- Pulling the glider manually instead of allowing it to track naturally.
- Bypassing or tampering with the locking mechanism.
- Carrying loads that prevent safe ladder climbing.
- **Descending too quickly, causing unnecessary shock loading on the system.**

3.5. MANTALING AND DISMANTLING A REMOVEABLE GLIDER

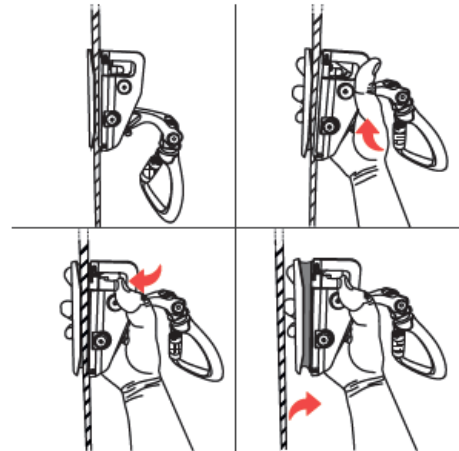
ASSEMBLY AND DISMANTLING

Assembly:

Rotate the cam upwards and introduce the Gravity glider on the wire cable, the security will automatically be opened by pushing the Gravity glider on the wire cable.

Dismantling:

Rotate the cam upwards, push the trigger to open the security and take the Gravity glider out of the line.



4. LIMITS OF USE

- The total weight of the user, including tools and equipment, must not exceed 150kg.
- The total weight of the user, tools and equipment not included, must not be less than 60kg.
- Before removing the glider from the lifeline or before detaching from the fall arrest system, fall protection must be provided by another device.
- The lifeline cannot be used as a work positioning system, in which case a separate restraint system must be used.
- Handling of the glider while climbing or descending may interfere with the braking mechanism. If necessary, handle the glider by its connector, never take in hand the body of the glider in service.
- **The user is not protected against the impact on the ground during the first 2 meters. Additional precautions must be taken when climbing or descending.**
- **In case of emergency the lifeline can be used to guide the descent to the ground of the person.**

- The glider including the lifeline should not be installed in a highly corrosive atmosphere (e.g. above a swimming pool) because of the risk of **non-visible stress corrosion cracking**, unless specific control measures are in place or compatibility is established.

5. PROHIBITED USE:

- The equipment stored in an environment that is aggressive to materials.
- Exceed the maximum number of users.
- **Only one user who works at height** and who is not under the control of a second person who can help him in case of need.
- The glider is intended for a single user, who is responsible for it. Use by more than one person is prohibited.
- The fall clearance is not sufficient.
- **The installation a lanyard between the absorber permanently mounted on the fall arrester and the harness, which** will increase the height of fall and eventually the fall will not be stopped.
- **The use of a lanyard that would allow the user to work completely off-centre compared to the system. Due to the additional lanyard the user will fall with the pendulum effect. Fall may not be stopped.**
- **If the fall arrester is equipped** with an energy absorber that is deployed or has a length of more than 10cm.
- The use of an additional piece of safety equipment in combination with that provided in the fall arrest system may be detrimental to safety.
- Using another energy absorber than the one originally fixed to fall arrest.
- When there is a risk of electric shock.
- In the event of frost or risk of frost, thunderstorms, or any other potentially dangerous exceptional weather conditions.
- It is forbidden to use the system as earthing of welding machines.
- If any doubt arises about its condition for safe use.
- Any use other than that intended by the manufacturer is prohibited.
- If the equipment has been used to arrest a fall and not used again until confirmed in writing by a competent person.